VIT-AP UNIVERSITY, ANDHRA PRADESH

Lab Sheet 7: MongoDB with Fronend

Branch/ Class: B.Tech/M.Tech **Date:** 05-03-2026
Faculty Name: Prof. S.Gopikrishnan **School:** SCOPE
Student name: Jahnavi Naidu **Reg. no.:** 23BCE8395

Question 1: Student Notes Manager (CRUD Operations)

Develop a simple web application that allows students to create, view, update, and delete study notes stored in MongoDB.

```
C:\Program Files\mongosh-2.7.0-win32-x64\mongosh-2.7.0-win32-x64\bin>cd bin
The system cannot find the path specified.

C:\Program Files\mongosh-2.7.0-win32-x64\mongosh-2.7.0-win32-x64\bin>mongosh
Current Mongosh Log ID: 69a928f7d2b2b51d6f7c2906
Connecting to:          mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000&appName=mongosh+2.7.0
Using MongoDB:          8.2.5
Using Mongosh:          2.7.0

For mongosh info see: https://www.mongodb.com/docs/mongodb-shell/

------
   The server generated these startup warnings when booting
   2026-02-28T19:45:37.131+05:30: Access control is not enabled for the database. Read and write access to data and configuration is
unrestricted
------

Enterprise test>
```

Modules

1. Add Note
2. View Notes
3. Edit Note
4. Delete Note

Functional Requirements

1. Add Note

Student enters:

- Title
- Subject
- Description

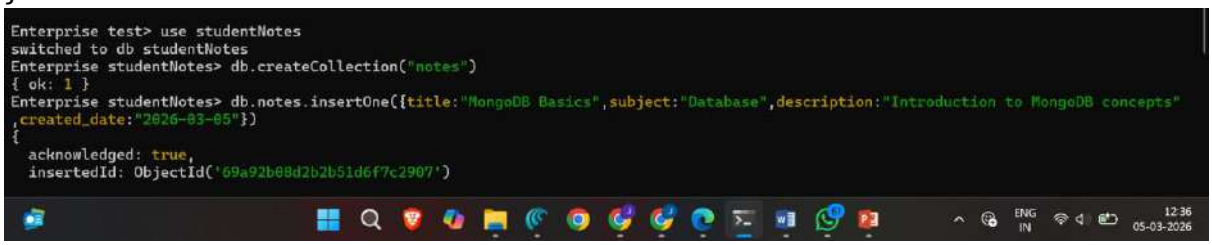
AJAX sends JSON data to server.
Example JSON request:

```
{
   "title":"MongoDB Basics",
   "subject":"Database",
   "description":"Introduction to MongoDB concepts"
   }
```

MongoDB stores document in **notes collection**.
Example MongoDB document:

```
{
   "_id":"ObjectId",
   "title":"MongoDB Basics",
   "subject":"Database",
   "description":"Introduction to MongoDB concepts",
   "created_date":"2026-03-05"
```

}



2. View Notes

Display all notes stored in database.

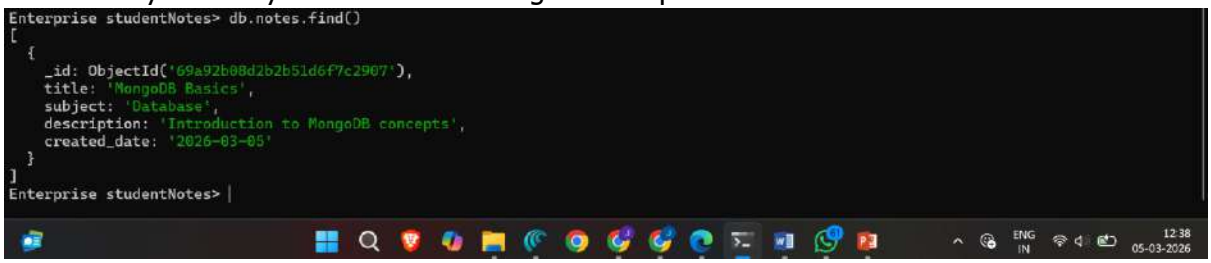
AJAX request:

GET /notes

Server returns JSON:

```
[
  {
  "title":"MongoDB Basics",
  "subject":"Database",
  "description":"Introduction to MongoDB concepts"
  }
  ]
```

Frontend dynamically renders notes using JavaScript.



3. Update Note

Student can modify title or description.

AJAX request:

PUT /notes/{id}

```
  {
  "title":"MongoDB Advanced",
  "description":"Aggregation and indexing"
  }
```

MongoDB operation:

```
db.notes.updateOne(
  {_id:ObjectId(id)},
```

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{$set:{title:"MongoDB Advanced",description:"Aggregation and indexing"}}
)

```
Enterprise studentNotes> db.notes.find()
[
  {
    _id: ObjectId('69a92b08d2b2b51d6f7c2907'),
    title: 'MongoDB Basics',
    subject: 'Database',
    description: 'Introduction to MongoDB concepts',
    created_date: '2026-03-05'
  }
]
Enterprise studentNotes> db.notes.updateOne( {_id:ObjectId("69a92b08d2b2b51d6f7c2907")}, {$set:{title:"MongoDB Advanced",description:
"Aggregation and indexing"}} )
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
Enterprise studentNotes> db.notes.find()
[
  {
    _id: ObjectId('69a92b08d2b2b51d6f7c2907'),
    title: 'MongoDB Advanced',
    subject: 'Database',
    description: 'Aggregation and indexing',
    created_date: '2026-03-05'
  }
]
Enterprise studentNotes>
```

4. Delete Note

AJAX request:
DELETE /notes/{id}
MongoDB query:
db.notes.deleteOne({_id:ObjectId(id)})

```
Enterprise studentNotes> db.notes.find()
[
  {
    _id: ObjectId('69a92b08d2b2b51d6f7c2907'),
    title: 'MongoDB Advanced',
    subject: 'Database',
    description: 'Aggregation and indexing',
    created_date: '2026-03-05'
  }
]
Enterprise studentNotes> db.notes.deleteOne({_id:ObjectId("69a92b08d2b2b51d6f7c2907")})
{ acknowledged: true, deletedCount: 1 }
Enterprise studentNotes> db.notes.find()

Enterprise studentNotes>
```

Question 2: Online Book Finder (Retrieval & Query Operations)

Develop a system where users can search, filter, and retrieve books using different MongoDB queries.

Modules

1. Search Books by Title
2. Filter Books by Category

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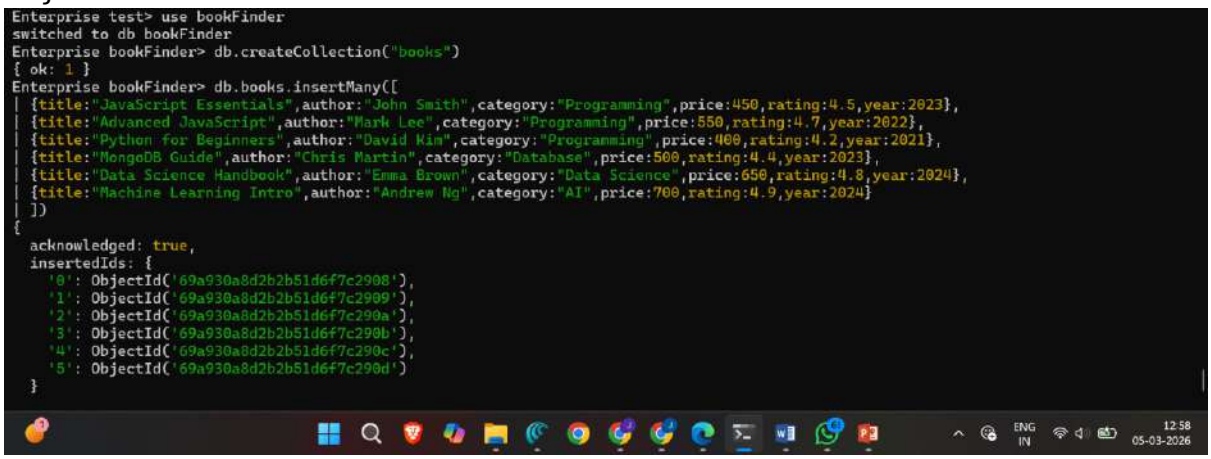
3. Sort Books by Price or Rating
4. Display Top Rated Books
5. Pagination (Load More)

Database Structure

Collection: **books**

Example document:

```
{
  "_id":"ObjectId",
  "title":"JavaScript Essentials",
  "author":"John Smith",
  "category":"Programming",
  "price":450,
  "rating":4.5,
  "year":2023
  }
```



Functional Requirements

1. Search Books by Title

AJAX request:

GET /books/search?title=javascript

MongoDB query:

db.books.find({title:{$regex:"javascript",$options:"i"}})

Returns JSON results.

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```
Enterprise bookFinder> db.books.find({title:{$regex:"javascript",$options:"i"}})
[
  {
    _id: ObjectId('69a930a8d2b2b51d6f7c2908'),
    title: 'JavaScript Essentials',
    author: 'John Smith',
    category: 'Programming',
    price: 450,
    rating: 4.5,
    year: 2023
  },
  {
    _id: ObjectId('69a930a8d2b2b51d6f7c2909'),
    title: 'Advanced JavaScript',
    author: 'Mark Lee',
    category: 'Programming',
    price: 550,
    rating: 4.7,
    year: 2022
  }
]
Enterprise bookFinder>
```

2. Filter Books by Category

AJAX request:

GET /books/category/programming

MongoDB query:

db.books.find({category:"Programming"})

```
Enterprise bookFinder> db.books.find({category:"Programming"})
[
  {
    _id: ObjectId('69a930a8d2b2b51d6f7c2908'),
    title: 'JavaScript Essentials',
    author: 'John Smith',
    category: 'Programming',
    price: 450,
    rating: 4.5,
    year: 2023
  },
  {
    _id: ObjectId('69a930a8d2b2b51d6f7c2909'),
    title: 'Advanced JavaScript',
    author: 'Mark Lee',
    category: 'Programming',
    price: 550,
    rating: 4.7,
    year: 2022
  },
  {
    _id: ObjectId('69a930a8d2b2b51d6f7c290a'),
    title: 'Python for Beginners',
    author: 'David Kim',
    category: 'Programming',
    price: 400,
    rating: 4.2,
    year: 2021
  }
]
Enterprise bookFinder>
```

3. Sort Books

Sort by price.

AJAX request:

GET /books/sort/price

MongoDB query:

db.books.find().sort({price:1})

Sort by rating.

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---

db.books.find().sort({rating:-1})

ASCENDING

```
Enterprise bookFinder> db.books.find().sort({price:1})
[
  {
    _id: ObjectId('69a930a8d2b2b51d6f7c290a'),
    title: 'Python for Beginners',
    author: 'David Kim',
    category: 'Programming',
    price: 400,
    rating: 4.2,
    year: 2021
  },
  {
    _id: ObjectId('69a930a8d2b2b51d6f7c2908'),
    title: 'JavaScript Essentials',
    author: 'John Smith',
    category: 'Programming',
    price: 450,
    rating: 4.5,
    year: 2023
  },
  {
    _id: ObjectId('69a930a8d2b2b51d6f7c290b'),
    title: 'MongoDB Guide',
    author: 'Chris Martin',
    category: 'Database',
    price: 500,
    rating: 4.4,
    year: 2023
  },
  {
    _id: ObjectId('69a930a8d2b2b51d6f7c2909'),
```

DESCENDING

```
Enterprise bookFinder> db.books.find().sort({rating:-1})
[
  {
    _id: ObjectId('69a930a8d2b2b51d6f7c290d'),
    title: 'Machine Learning Intro',
    author: 'Andrew Ng',
    category: 'AI',
    price: 700,
    rating: 4.9,
    year: 2024
  },
  {
    _id: ObjectId('69a930a8d2b2b51d6f7c290c'),
    title: 'Data Science Handbook',
    author: 'Emma Brown',
    category: 'Data Science',
    price: 650,
    rating: 4.8,
    year: 2024
  },
  {
    _id: ObjectId('69a930a8d2b2b51d6f7c2909'),
    title: 'Advanced JavaScript',
    author: 'Mark Lee',
    category: 'Programming',
    price: 550,
    rating: 4.7,
    year: 2022
  },
  {
    _id: ObjectId('69a930a8d2b2b51d6f7c2908'),
```

4. Top Rated Books

AJAX request:
GET /books/top
MongoDB query:
db.books.find({rating:{$gte:4}}).limit(5)

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```
Enterprise bookFinder> db.books.find({rating:{$gte:4}}).limit(5)
[
  {
    _id: ObjectId('69a930a8d2b2b51d6f7c2908'),
    title: 'JavaScript Essentials',
    author: 'John Smith',
    category: 'Programming',
    price: 450,
    rating: 4.5,
    year: 2023
  },
  {
    _id: ObjectId('69a930a8d2b2b51d6f7c2909'),
    title: 'Advanced JavaScript',
    author: 'Mark Lee',
    category: 'Programming',
    price: 550,
    rating: 4.7,
    year: 2022
  },
  {
    _id: ObjectId('69a930a8d2b2b51d6f7c290a'),
    title: 'Python for Beginners',
    author: 'David Kim',
    category: 'Programming',
    price: 400,
    rating: 4.2,
    year: 2021
  },
  {
    _id: ObjectId('69a930a8d2b2b51d6f7c290b'),
```

5. Pagination

AJAX request:

GET /books?page=2

MongoDB query:

```
db.books.find()
    .skip(5)
    .limit(5)
```


```
Enterprise bookFinder> db.books.find({rating:{$gte:4}}).limit(5)
[
  {
    _id: ObjectId('69a930a8d2b2b51d6f7c2908'),
    title: 'JavaScript Essentials',
    author: 'John Smith',
    category: 'Programming',
    price: 450,
    rating: 4.5,
    year: 2023
  },
  {
    _id: ObjectId('69a930a8d2b2b51d6f7c2909'),
    title: 'Advanced JavaScript',
    author: 'Mark Lee',
    category: 'Programming',
    price: 550,
    rating: 4.7,
    year: 2022
  },
  {
    _id: ObjectId('69a930a8d2b2b51d6f7c290a'),
    title: 'Python for Beginners',
    author: 'David Kim',
    category: 'Programming',
    price: 400,
    rating: 4.2,
    year: 2021
  },
  {
    _id: ObjectId('69a930a8d2b2b51d6f7c290b'),
```